

# The Fermi–Pasta–Ulam–Tsingou Problem

## Nonlinear Dynamics and Recurrence Phenomena

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### Introduction

The Fermi–Pasta–Ulam–Tsingou (FPUT) problem is a paradox in physics that originated from a numerical experiment at Los Alamos National Laboratory.[1] It followed from an experiment to study how energy spreads in a non-linear system of connected masses. Surprisingly, instead of the energy dispersing evenly, as statistical mechanics predicted, it repeatedly returned to its initial state—a phenomenon now known as FPUT recurrence. This unexpected result challenged the assumption that all non-linear systems lead to thermal equilibrium, marking a key moment in the birth of nonlinear science, chaos theory, and computational physics.[5], [7]

This poster investigates the FPUT problem by briefly reviewing its historical motivation, presenting the non-linear equations that define the system, and analysing results from numerical simulations. We focus on the unexpected energy recurrence and its broader connections to solitons, chaos, and the foundations of non-linear physics.[2], [4]

### Mathematical Model

The FPUT system describes a one-dimensional chain of  $N$  identical particles connected in a chain of non-linear springs. It was designed to test the effects of small non-linearities on energy distribution among vibrational modes.[1]

Two versions of the model are as follows:

**FPUT- $\alpha$  model** (quadratic non-linearity):

$$m\ddot{x}_n = k(x_{n+1} - x_n) - k(x_n - x_{n-1}) + \alpha \left[ (x_{n+1} - x_n)^2 - (x_n - x_{n-1})^2 \right] \quad (1)$$

**FPUT- $\beta$  model** (cubic non-linearity):

$$m\ddot{x}_n = k(x_{n+1} - x_n) - k(x_n - x_{n-1}) + \beta \left[ (x_{n+1} - x_n)^3 - (x_n - x_{n-1})^3 \right] \quad (2)$$

where  $x_n$  is the displacement of the  $n$ -th mass from equilibrium. The parameters  $\alpha$  and  $\beta$  determine the strength of the non-linear interaction. The linear terms alone describes a harmonic chain, while the non-linear terms enable energy transfer between modes.[4]

### Initial Conditions

At the start of the simulation, the system is set up by exciting just a single vibrational mode—usually the lowest one (Mode 1)—while leaving all others inactive. The initial displacements follow a sine wave pattern:[1]

$$x_n(0) = A \sin\left(\frac{\pi n}{N+1}\right), \quad \dot{x}_n(0) = 0$$

This means each mass starts at rest, but with a position offset that matches the shape of the first mode.

The ends of the chain are held fixed with boundary conditions:

$$x_0 = x_{N+1} = 0$$

This setup makes it easy to see how energy moves through the system over time, and helps highlight any unexpected behavior—like the recurrence effect discovered in the original FPUT study.[1], [2]

### Numerical Simulation

To study how the system evolves over time, the equations of motion are solved numerically. Because the FPUT system is nonlinear and lacks an analytical solution, we rely on time-stepping algorithms to approximate its dynamics.

We use a symplectic integrator—in this case, the velocity Verlet method—which is well-suited for conserving total energy in Hamiltonian systems.[4] This allows us to track how energy shifts between modes without introducing artificial damping or drift.

The simulation was implemented in Python using NumPy for efficient array computations. The system consisted of  $N = 32$  particles with a time step of  $\Delta t = 0.01$ , evolved over 50,000 steps. Both  $\alpha$  and  $\beta$  parameters were varied to explore different nonlinear regimes, with typical values in the range  $0.1 \leq \alpha, \beta \leq 1.0$ .

These settings strike a balance between numerical stability and the ability to observe long-term behavior, such as energy spreading and recurrence.[2]

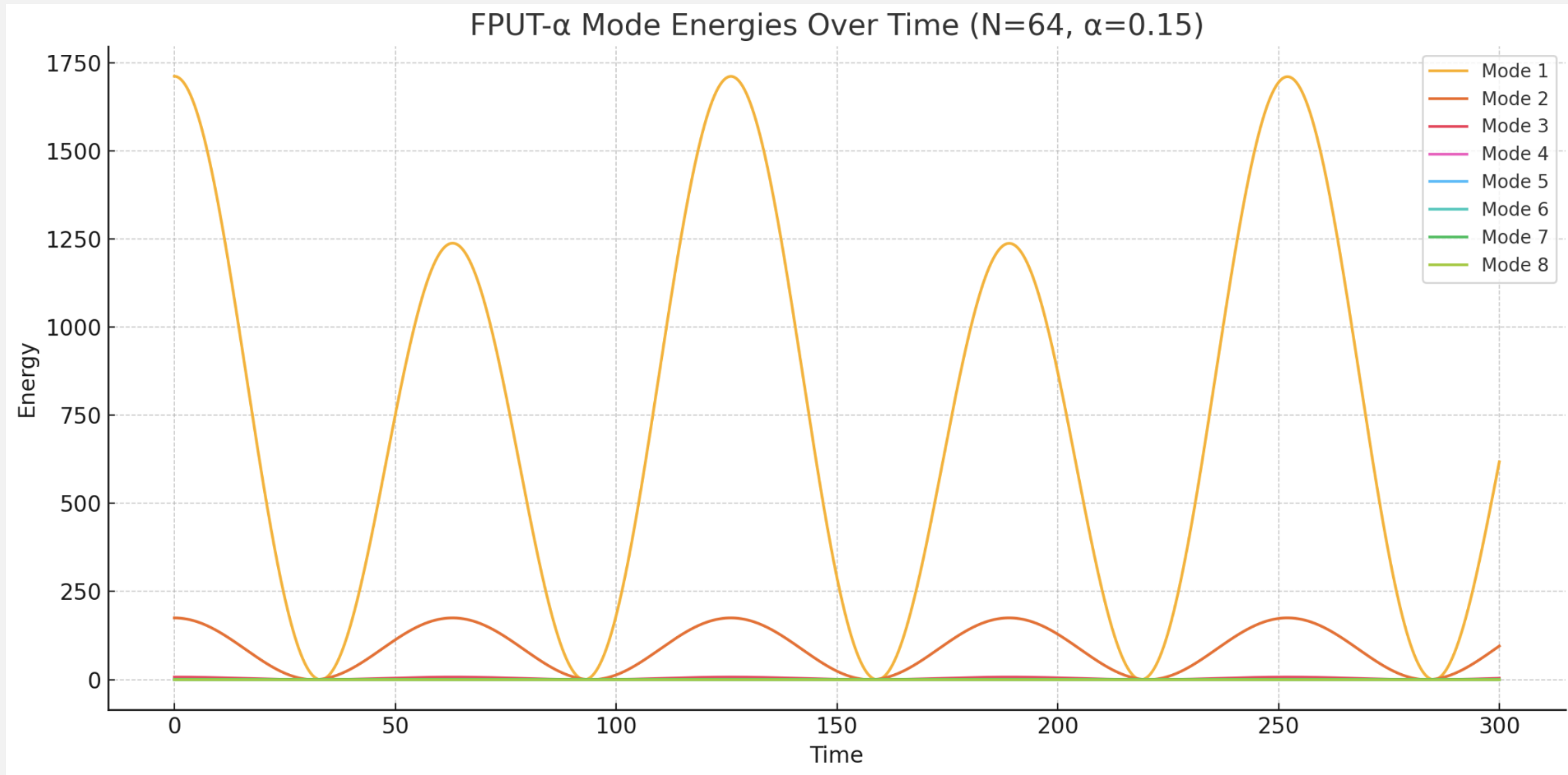
### Results and Plots

The simulation results reveal how energy evolves across the normal modes of the system over time. Starting with all energy in the first mode, we observe that it gradually spreads to higher modes due to the system’s non-linear interactions.

Interestingly, instead of reaching a state of thermal equilibrium, the energy partially returns to the initial mode—a behavior known as FPUT recurrence.[1], [2] This recurrence is especially visible when non-linearities are weak, and becomes less regular as the system becomes more chaotic at higher  $\alpha$  or  $\beta$  values.

The plots below show the time evolution of modal energies. The first mode’s energy decreases, redistributes, and then unexpectedly recovers, confirming the recurrence effect first reported by Fermi, Pasta, Ulam, and Tsingou.

### Mode Energy Plot



Energy of the first few modes over time for the FPUT- $\alpha$  model.

### Key Observations

The simulation confirms several of the key behaviors first reported in the original FPUT experiment.[1] Most notably, we observe that energy does not simply spread out and remain evenly distributed among all modes as classical statistical mechanics would predict.

Instead, after an initial redistribution, the system shows a partial return of energy to the first mode—this is the recurrence effect that challenged early assumptions about thermalisation [2], [5] The effect is strongest when the non-linear terms are small, and becomes less regular as the non-linearity increases.

We also see that higher modes gain energy temporarily but never dominate the dynamics. Overall, the energy exchange remains highly structured and reveals a kind of hidden order, even in a non-linear system.

### Modern Connections

The FPUT problem, while originally designed to test ideas in statistical mechanics, ended up reshaping how we understand non-linear systems. Its surprising results directly inspired the discovery of solitons and the Korteweg–de Vries (KdV) equation[2], which models shallow water waves and other wave-like phenomena in nonlinear media.

Beyond soliton theory, FPUT opened the door to the study of chaos, integrable systems, and long-term stability in Hamiltonian dynamics. It also played a major role in the emergence of computational physics,[7] showing that computers could reveal physical insights not accessible through pure mathematics.

Today, ideas rooted in FPUT dynamics appear in fields as diverse as condensed matter physics, optical fiber communication, lattice vibrations, and even neuroscience. What started as a puzzling numerical experiment is now seen as a foundational moment in the study of complexity and order in non-linear systems.[4]

### Conclusion

The FPUT problem started as a simple test of how energy spreads in a nonlinear system. But the unexpected recurrence effect revealed that even small nonlinearities can lead to rich and structured behavior—behavior that defies our basic expectations about thermalisation.

Through simulation, we’ve seen how energy moves between modes in a way that’s both predictable and surprising. These dynamics not only challenge classical assumptions, but also highlight the importance of computational approaches in modern physics.

Even decades later, the FPUT system continues to offer insights into how order and complexity emerge from simple rules—making it as relevant today as it was when it was first run on one of the world’s earliest computers.

### References

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